# Predictive vs Generative

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

## Teach generation as prediction under an objective, not an automatic fact check.

Supply complete count tables and task cards, keep cases separate and assess the fresh signal example.

**Keep one habit:** Require a conditioning prefix and the appropriate evaluation evidence beside a probability or factual claim.

MAIN LESSON · SECTION 1

## Prepare this connected week

> Use one main plan and keep practice separate from the independent first attempt.

### Learning goal and precision

Primary target: Distinguish discriminative **prediction** from generation by inspecting the requested **output**

Use two 45-minute sessions. All model **output**s and probability tables are constructed classroom examples. No hosted AI service, external API or student account is needed.

Distinguish a model objective from factual support and system permissions. Collect independent work before discussing its solutions.

By the end, I can distinguish a generated continuation from a **prediction** checked against the world.

## What to prepare

- Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.
- Case exposure check: this source example appears in ai-2/27, ai-3/1. Reuse a familiar case for recall or modelling; verify that the case used to assess transfer has not already received feedback.
- Printed lesson and workbook; paper and pencil
- Calculator; optional local Python 3 interpreter
- Prepare the independent task and fresh retry without displaying solutions

## Main teaching plan

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

## Opening retrieval and readiness

Without opening the old lesson, explain one key idea from Image and Voice Pipelines. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of speech pipelines and multimodal **evidence**

Reference example, not the only acceptable wording: Teacher explanation: A multimodal pipeline may transcribe speech, inspect an image and generate a response.

Errors at one stage can propagate even when later stages run correctly. Track source modality, segment/time references, transcription, interpretation and final claims separately.

Compare conflicting **evidence** without assuming one modality always wins. Ask for clarification when the task needs unresolved information.

Without opening the old lesson, explain one key idea from Feedback Loops. Give an example and a mistake that the idea helps you avoid.

Look for: Implement a deterministic recommender

Reference example, not the only acceptable wording: Both score 8 and qualify; A is alphabetically first under the tie rule.

Without opening the old lesson, explain one key idea from Similarity by Selected Traits. Give an example and a mistake that the idea helps you avoid.

Look for: Represent items with numeric features

Reference example, not the only acceptable wording: Energy 4, group work 1.

A **prediction** is a model **output**. Truth requires appropriate independent **observation**s or sources; confidence and low **training** loss do not supply that **evidence**.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

### Teach the learning cycle

### Retrieval answer guidance

### Image and Voice Pipelines

### Feedback Loops

### Similarity by Selected Traits

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

### What the learner must demonstrate

A tool retrieves a book description, predicts a genre, then writes a new blurb. Identify each operation and give one distinct failure to check at each stage.

### Concept to explain

**Predictive** systems select a target; **generative** systems construct new **output**.
</details>

| Term | Meaning |
| --- | --- |
| predictive | Estimating a target from supplied inputs. |
| generative | Producing new content from learned patterns and supplied conditions. |
| output space | The set or range of outputs a system can produce for a task. |
| fit for purpose | Suitable for the stated users, task, conditions and consequences. |

### Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

### Preparation for the connected explanation

### Readiness answer

### End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

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MAIN LESSON · SECTION 2

## Teach objectives and empirical probabilities without overstating truth

> Keep marker, door, box and signal cases separately labelled.

### State component targets and next-token-fit limits

A classifier might map a message to the label request or report. A text generator instead produces a sequence, predicting a distribution over the next token given a prefix.

Both make **prediction**s, but the **training** targets and permitted **output**s differ. One application can combine generation with a classifier, retrieval and human approval.

Calling a whole product "**generative**" does not describe every component.

The objective defines what **training** attempts to fit. Next-token fit rewards matching observed continuations; it does not directly certify that an answer is true, well sourced or permitted.

A fluent completion is still an **output** to evaluate against the task.

### Supply and compute the marker table

| Prefix | Continuation | Count |
|---|---|---|
| The marker is | red | 6 |
| The marker is | blue | 3 |
| The marker is | green | 1 |

There are ten observed continuations. The empirical probabilities are red 6/10 = 0.6, blue 3/10 = 0.3, green 1/10 = 0.1. They are nonnegative and sum to 1.

A simple frequentist classroom model uses these relative counts as its next-token distribution. These probabilities describe the supplied corpus counts, not the current colour of any actual marker.

## Expand the prefix before choosing its next alternatives

After producing red, the next step conditions on the expanded prefix "The marker is red". A second table might give an end token probability 0.8 and the token today probability 0.2.

A generated sequence can stop at the end token or continue. Do not reuse the first-step table unchanged after adding a token unless the model explicitly ignores the new prefix.

## Contrast factual evaluation with continuation

A separate classifier can return red as a label from an image; that is a different **input** and target from continuing text. A downstream evaluation asks whether the system correctly reports the current marker.

It must include ground truth or sources and appropriate failure handling, not just count plausible phrases.

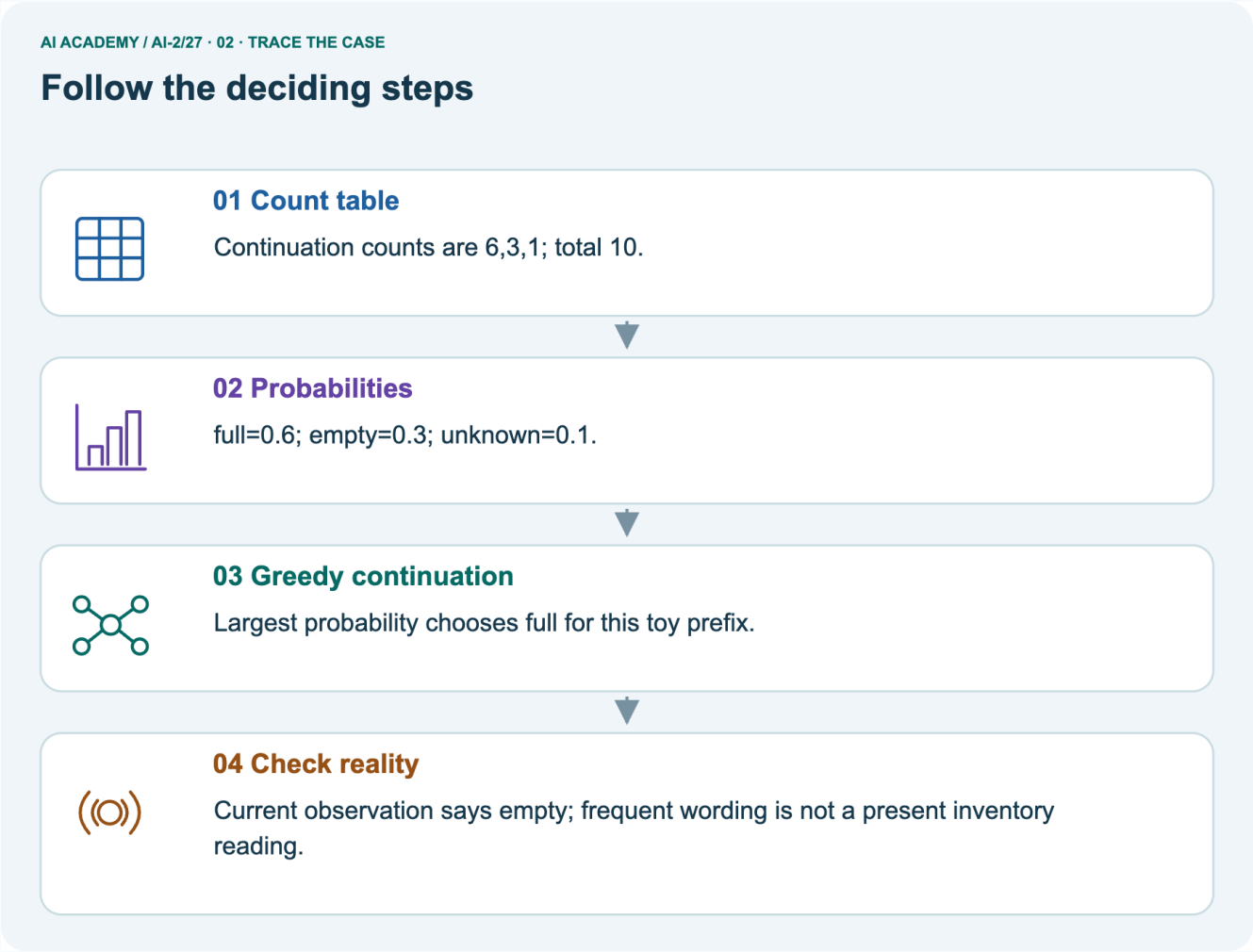
## Read the separate box corpus and current observation

A fictional toy generator predicts the next word after “The box is”.

Observed **training** continuations: full 6, empty 3, unknown 1.

Use counts divided by total 10. The current physical box is separately checked and found empty.

## Trace greedy output and independent evidence




Follow the labelled picture with the matching explanation. Open the vector diagram

The table predicts a continuation in a text exercise, not the contents of the current box. Generating the most frequent token is a model-defined decision. A factual inventory statement needs current **evidence**.

The numerical table is intentionally small; real **generative** systems condition on richer context and do not simply consult these three counts.

A one-step toy distribution does not describe every **predictive** or **generative** model or validate factual accuracy.

## Change the corpus without changing the box

Change: Change the **training** table to full 2, empty 7, unknown 1.

**Prediction**: Greedy **output** becomes empty; the model assigns that token probability 0.7.

Reason: The continuation distribution changed; the independently observed box has not changed.

## Elicit the evidence-limited explanation

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

### ▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

**Input** → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. ‘Is this book suitable?’ → classification.

2. ‘How many days to finish?’ → numeric **prediction**.

3. ‘Write a spoiler-free teaser’ → generation.

4. A product may combine them, but each stage needs its own test.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

#### Concrete teaching case

A labels genre; B writes a new blurb; C computes 24/6.

Distinguish operations.

### Exact explanation to model

A predicts a category, B generates text, C calculates 4. Products may combine them.

### Explain the added worked example

### Key terms

| Term | Meaning |
|---|---|
| Generative objective | A training aim concerned with producing or modelling possible outputs |
| Discriminative objective | A training aim mapping inputs to a specified target such as a label |
| Conditional distribution | Probabilities for alternatives given stated information |
| Autoregressive generation | Producing a sequence one next item at a time using preceding items |

### Model the reasoning aloud

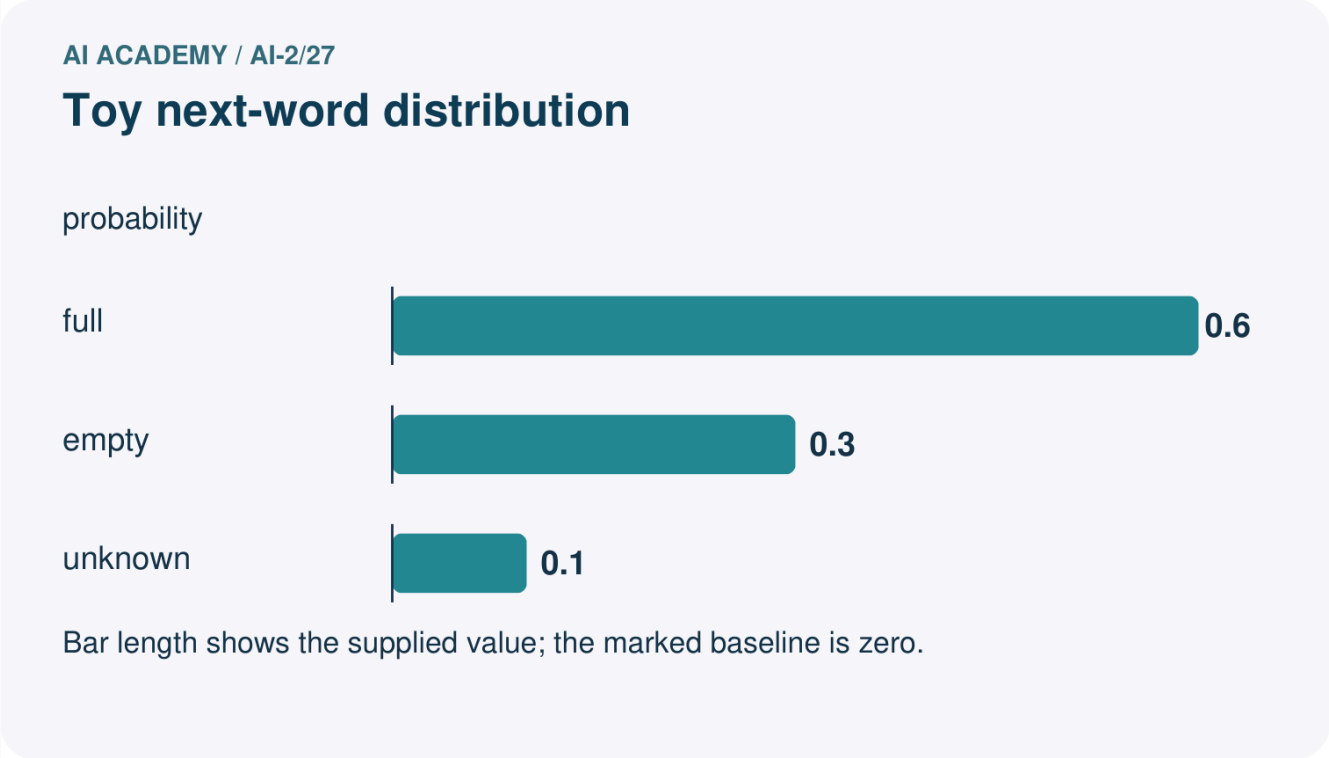
Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

| Thinking move | Teacher prompt |
|---|---|
| Plan | What is given? What is the question? Which procedure fits these facts? |
| Do | Point to each input and intermediate step in the worked trace. Name the rule that allows the next step. |
| Check | Does the output answer the question? Which assumption or limitation prevents a larger claim? |

## A pictorial worked case: Predictive vs Generative

### The facts we will use




Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/27

# Predictive vs Generative

**Start with the supplied facts**

Follow each arrow. Ask which fact allows the next step.



### 1. Count table

Continuation counts are 6,3,1; total 10.



### 2. Probabilities

full=0.6; empty=0.3; unknown=0.1.



### 3. Greedy continuation

Largest probability chooses full for this toy prefix.



### 4. Check reality

Current observation says empty; frequent wording is not a present inventory reading.

**CHANGE ONE THING**

Change the training table to full 2, empty 7, unknown 1.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

### Read the picture one step at a time

| STEP | WHAT HAPPENS AND WHY |
| --- | --- |
| 1. Count table | Continuation counts are 6,3,1; total 10. |
| 2. Probabilities | full=0.6; empty=0.3; unknown=0.1. |
| 3. Greedy continuation | Largest probability chooses full for this toy prefix. |
| 4. Check reality | Current observation says empty; frequent wording is not a present inventory reading. |

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

### Why this result follows

### What this example does not establish

### Change one thing in the pictured case

### Ask while showing the picture

# Guide the reasoning, then remove help

> Collect a response from each learner before discussing the worked key.

## Prompt before revealing

Completion task: For The marker is, calculate the three empirical probabilities, their sum and their meaning.

## Expected reasoning and fading help

Teacher check: Teacher explanation: Probabilities 0.6,0.3,0.1 sum to 1 and describe ten supplied corpus continuations; they are not **observation**s of the current marker.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

## Check the misconception before discussion

Use the worked case for **Generative** and **predictive** objectives. Proposed explanation: "**Generative** models do not predict". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: **Autoregressive generation** predicts the next item repeatedly.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

## Use the pictorial check

Does matching the physical box in the changed table prove the generator measures boxes?

No. It happens to match here; its **mechanism** still selects a token from supplied counts rather than sensing contents.

Changed-case reasoning: Greedy **output** becomes empty; the model assigns that token probability 0.7. The continuation distribution changed; the independently observed box has not changed.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

## Keep each learner explaining

- Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.
- Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.
- In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Demonstrate and guide practice

### Board sequence

Write the given facts: A labels genre; B writes a new blurb; C computes 24/6.

Write the question to explain: Distinguish operations.

Reveal after **prediction**s: A predicts a category, B generates text, C calculates 4. Products may combine them.

Use the earlier changed question as optional later practice. Reserve Analyse a new corpus independently for the first independent attempt before hints or feedback.
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### Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

### Model these guided questions

Guided: open 0.8 and closed 0.2. The corpus prefers open, but the sensor supplies current closed **evidence**. Exit: a conditional continuation probability is not verification of a current fact.

### Case-specific teaching sequence

| Stage | Teacher action |
|---|---|
| Session 1 start to 7 minutes | Retrieve the prerequisite and distinguish a numerical claim from its interpretation. |
| 7 to 18 minutes | Read section 1 and complete W1; ask for a reason alongside each answer. |
| 18 to 32 minutes | Trace the worked case in section 2 and complete W2. |
| 32 to 45 minutes | Discuss the diagram and complete W3 in pairs; record support. |
| Session 2 start to 8 minutes | Complete the guided response before sharing the answer. |
| 8 to 20 minutes | Read the remaining sections and trace another constructed case. |
| 20 to 35 minutes | Collect W4 independently, requiring calculations and a bounded conclusion. |
| 35 to 45 minutes | Discuss W5 and the exit response. Provide feedback and a fresh retry. |

### Reduce help deliberately

### Individual reasoning check

### Pictorial practice question

### Reasoned teacher answer

# Run the supplied investigation

> Keep expected results, observed paper traces and remaining limits distinguishable.

## Supplied materials and answer guidance

Marker6/10,3/10,1/10→0.6,0.3,0.1,sum1; next-prefix end/today0.8/0.2 is not a replacement colour distribution. Door4/5open,1/5closed still conflicts with supplied current closed if greedyopen. Separate box counts6/3/1 greedyfull conflict with currentempty; changed2/7/1 greedilyempty does not imply measurement. Task cards distinguish stated calculation, retrieval, label/number prediction, sequence output and combined stages; no model performance is measured. Reserve the signal corpus for W4.

## Run and record

1. Predict → Test → Record → Explain
2. Keep expected and actual results separate. Surprises are useful data when recorded honestly.
3. Use the supplied nine component cards. State input/**output** contracts, distinguish the stated **mechanism**s and propose evidence-based checks. No real model **output**s or performance scores are supplied for these task cards.
4. What did the hardest test teach you that the normal test did not?

## Keep an inspectable record

| Record | What to capture |
| --- | --- |
| Input or evidence | Exact data, instruction or claim |
| Expected result | Answer or safe fallback declared before the run |

| Record | What to capture |
| --- | --- |
| Observed result | Actual result, including errors |
| Explanation | Rule, calculation or evidence explaining the outcome |
| Next test | One justified change and a new prediction |

## Diagnose the first misunderstanding

**Generative** models do not predict: **Autoregressive generation** predicts the next item repeatedly.

The most frequent phrase is the current truth: Corpus frequency and current **observation**s answer different questions.

A single accuracy score evaluates the whole system: Generation, retrieval, grounding, formatting and permissions require appropriate separate checks.

## Choose support from the response

| What the response shows | Next teaching action |
| --- | --- |
| Missing or misread facts | Read the case aloud, mark supplied versus unknown facts, and let the learner restate it. |
| Wrong procedure | Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation. |
| Arithmetic or implementation error | Trace intermediate values or lines; preserve the first mismatch and retry that step. |
| Correct result without a reason | Ask for a trace and a changed case before marking independent understanding. |

| What the response shows | Next teaching action |
| --- | --- |
| An unsupported wider claim | Ask which supplied evidence supports the claim and which further observation would be needed. |

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

### ▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

#### Make the lab runnable

Use the supplied investigation for shared practice. Keep predictions and observed paper-trace results separate. Reserve Analyse a new corpus independently for the individual first attempt before feedback; do not use the reserved case as a paired lab task.

#### Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

#### Address these additional misconceptions

#### Match the support to the first error

## Assess the selected new case

> Keep this key until after the first attempt; accept valid alternatives with defensible reasoning.

### Selected independent question and key

Selected case: For prefix The signal is, counts are high 2, low 5, unknown 3. Compute all probabilities and the most frequent continuation. A current measurement reports high.

State what the most frequent continuation does and does not establish.

Teacher explanation: Probabilities high 0.2, low 0.5, unknown 0.3; low is most frequent. This describes the constructed corpus, not the current signal.

Current high is supported by the supplied measurement; generating low would conflict with that **observation**.

## Judge explanation, trace and limits

| Dimension | Evidence required |
|---|---|
| Explanation | Names the applicable idea or procedure and ties it to the supplied facts. |
| Trace and result | Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution. |
| Limits | States the assumptions, missing evidence or conditions under which the result would change. |

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

## Give one usable repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

### W1 Identify the objective

Classify these components by target: label a message request/report; continue a prefix with a sequence; retrieve a stored sensor record. Explain why the application may combine them.

Teacher explanation: Message labels are discriminative **prediction**; continuing a sequence is generation; record lookup is retrieval rather than learned generation by itself.

One application can route a request, retrieve **evidence** and generate a response.

### W2 Compute the distribution

For The marker is, calculate the three empirical probabilities, their sum and their meaning.

Teacher explanation: Probabilities 0.6,0.3,0.1 sum to 1 and describe ten supplied corpus continuations; they are not **observation**s of the current marker.

### W3 Follow the prefix

After red is selected, the expanded-prefix table gives end 0.8 and today 0.2. Explain why these values cannot be substituted for the red/blue/green probabilities.

Teacher explanation: The conditional prefix and alternatives differ. The second table describes what follows The marker is red, not what colour follows the earlier prefix.

### W5 Design a useful evaluation

A system reports current room status using retrieved measurements and generated sentences. Name checks for **evidence**, response quality and permission.

Teacher explanation: Check the correct room and measurement time/source, agreement of every status claim with retrieved data, required response format and handling of missing/conflicting **observation**s, and that only approved read/report actions occur.

Continuation fit alone is insufficient.

## ▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

### Worked case answer

A predicts a category, B generates text, C calculates 4. Products may combine them.

### Additional transfer practice question

One model labels a review positive or negative; another writes a new review. Which classifies and which generates? Must generated text be true?

### Reasoned practice answer

The first classifies; the second generates. Generated text can be inaccurate or invented, so a fluent review is not verified **evidence**.

### What counts as evidence

| Criterion | 0 points | 1 point | 2 points |
|---|---|---|---|
| Result | Incorrect or absent | Partly correct | Correct with necessary conditions |
| Reasoning | No supporting trace | Some steps | Clear mechanism, calculation or source support |
| Limit | Overclaims certainty | Vague limitation | Relevant uncertainty or missing fact |
| Transfer | Copies without adapting | Adapts with prompts | Solves a changed case independently |

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

### Feedback

Ask the learner to find the **evidence** that challenges this claim: “Generating text proves the model checked the facts.” Keep the first attempt and an uncoached retry separate.

### Original transfer answer

Retrieval may fetch the wrong edition; **prediction** may choose the wrong genre; generation may invent an unsupported detail. Check document identity, the genre against reliable **evidence** and each factual claim in the blurb.

Fluency does not validate the preceding stages.

### Explanations for the additional workbook examples

### W1 Identify the objective

### W2 Compute the distribution

### W3 Follow the prefix

### W4 Analyse a new corpus independently

For prefix The signal is, counts are high 2, low 5, unknown 3. Compute all probabilities and the most frequent continuation. A current measurement reports high.

State what the most frequent continuation does and does not establish.

Teacher explanation: Probabilities high 0.2, low 0.5, unknown 0.3; low is most frequent. This describes the constructed corpus, not the current signal.

Current high is supported by the supplied measurement; generating low would conflict with that **observation**.

### W5 Design a useful evaluation

### Assess independent transfer

### Selected case answer

### Feedback that the learner uses

MAIN LESSON · SECTION 6

# Connect, adapt and return

> Retain the project evidence and revisit the reasoning on a changed case.

## Project evidence to keep

Keep scoring **predictive** and explanations templated or carefully generated and verified.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Use approved book facts for recommender explanations; generated fluency is not a replacement for catalogue **evidence**.

## Support while preserving the reasoning goal

Trace one record or condition at a time with counters and labelled columns. Stretch: introduce a tie, boundary or missing field and declare its handling policy before rerunning.

Provide labelled intermediate columns and allow a calculator. Ask students to state the meaning and denominator of a value before interpreting it.

Extension: implement the classroom calculation locally and test boundary cases without using personal data.

W4 requires 0.2,0.5,0.3 and low, with the corpus/current-**evidence** distinction. Fresh retry: counts ready 7, busy 1, unknown 2 and current **observation** busy.

Probabilities 0.7,0.1,0.2; ready is most frequent but does not establish the current state.

## Plan the delayed check

Suggested return points in this level: ai-2/28, ai-2/30, ai-2/33. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

| Record | Teacher evidence |
| --- | --- |
| Learner code and date | |
| Concept and case used; prior exposure | |
| Explanation / trace / limits | |
| Support and first repair | |
| Changed case and delayed outcome | |
| Next teaching action | |

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

## Bridge to next week

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 28: Prompt Experiments. Retrieve today's **evidence** and identify what carries forward and what is new.

Next is Week28: Prompt Experiments. Carry forward component objectives, fixed evidence and separate factual/task criteria before comparing prompt variants.

### ▼ Earlier wording, alternatives and picture transcripts

Connect and extend learning

Transparent Book Recommender

Support and stretch

Exit reflection

Bridge to the next week

Support and fresh reassessment

Plan the delayed return

Project evidence from this case

# Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference: alternative planning checklist**

REFERENCE · SECTION 7

## Reference: alternative planning checklist

> Use as a reminder alongside the main route rather than teaching the cycle twice.

### Preparation and schedule

Teaching target: Distinguish discriminative **prediction** from generation by inspecting the requested **output**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
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> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

## Practice and assessment reminders

Before class, solve the supported practice: For The marker is, calculate the three empirical probabilities, their sum and their meaning.

Reasoned practice answer: Teacher explanation: Probabilities 0.6,0.3,0.1 sum to 1 and describe ten supplied corpus continuations; they are not **observation**s of the current marker.

Check every learner before discussion: Use the worked case for **Generative** and **predictive** objectives. Proposed explanation: "**Generative** models do not predict". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: **Autoregressive generation** predicts the next item repeatedly.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Change the **training** table to full 2, empty 7, unknown 1. Require a **prediction** and a reason before comparing. Expected result: Greedy **output** becomes empty; the model assigns that token probability 0.7.

Reason: The continuation distribution changed; the independently observed box has not changed.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

| Criterion | Not yet | With support | Independent |
|---|---|---|---|
| Explanation | Names an answer without a deciding fact | Uses a hint to link fact and decision | Explains the deciding fact in own words |
| Trace or test | Cannot connect input and result | Completes steps with prompts | Traces or tests a changed case accurately |
| Limits and revision | Claims more than evidence supports | Names a limit after a prompt | States a defensible limit and useful next test |

Project checkpoint: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference: feedback and records**

REFERENCE · SECTION 8

# Reference: feedback and records

> Use digital tools where they help retain an explanation and its evidence.

## Keep practice and independent evidence separate

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

### Repair and portfolio records

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.
</details>

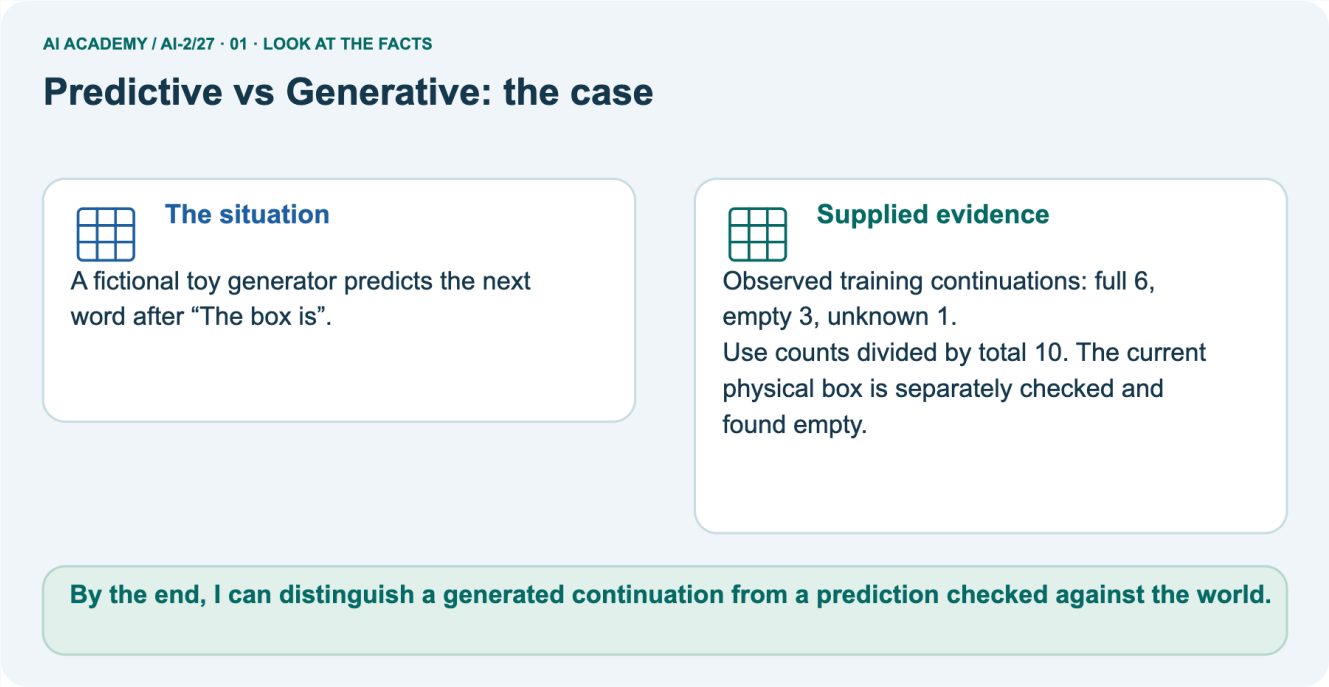
<details>
<summary>Reference: teach with the six pictures</summary>
</details>

REFERENCE · SECTION 9

## Reference: teach with the six pictures

> Choose the picture for the teaching move and connect it to the explanation.

## 01. Predictive vs Generative: the case




Follow the labelled picture with the matching explanation. Open the vector diagram

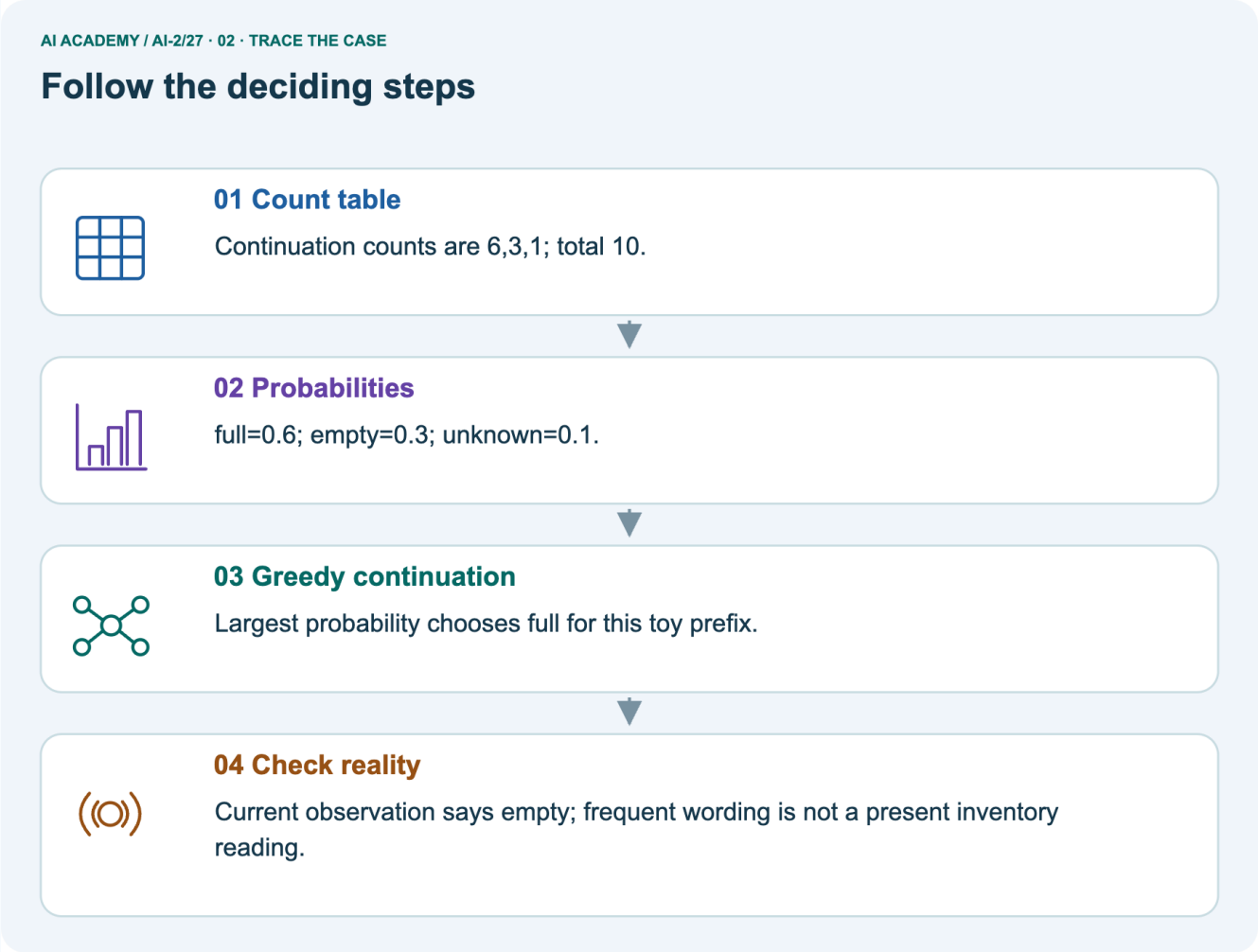
A fictional toy generator predicts the next word after “The box is”. Supplied facts: Observed **training** continuations: full 6, empty 3, unknown 1. Use counts divided by total 10.

The current physical box is separately checked and found empty.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

## 02. Follow the deciding steps



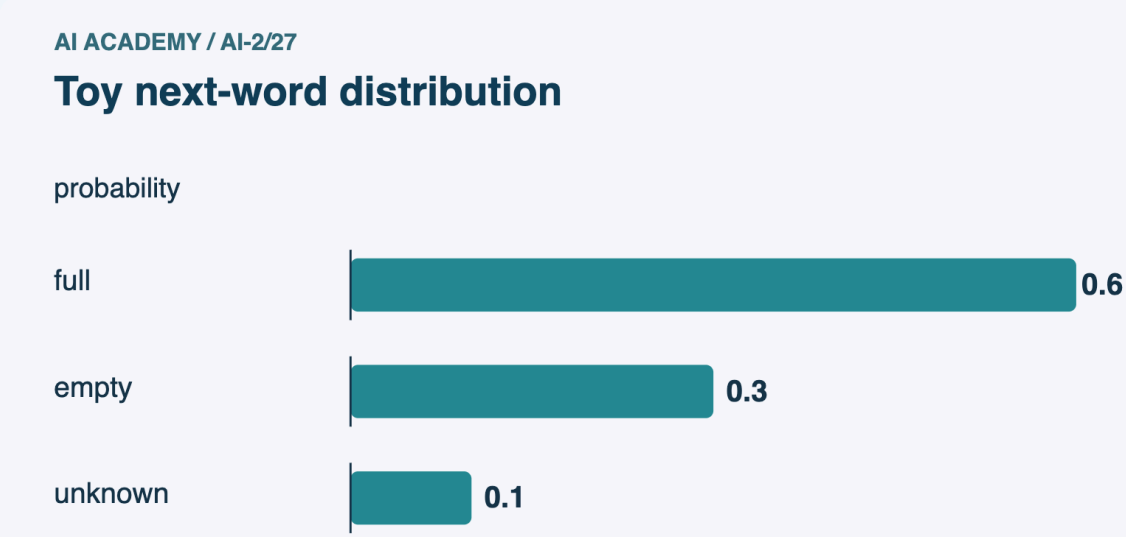

Follow the labelled picture with the matching explanation. Open the vector diagram

Count table: Continuation counts are 6,3,1; total 10. → Probabilities: full=0.6; empty=0.3; unknown=0.1. → Greedy continuation: Largest probability chooses full for this toy prefix. → Check reality: Current **observation** says empty; frequent wording is not a present inventory reading.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

## 03. Toy next-word distribution




Follow the labelled picture with the matching explanation. Open the vector diagram

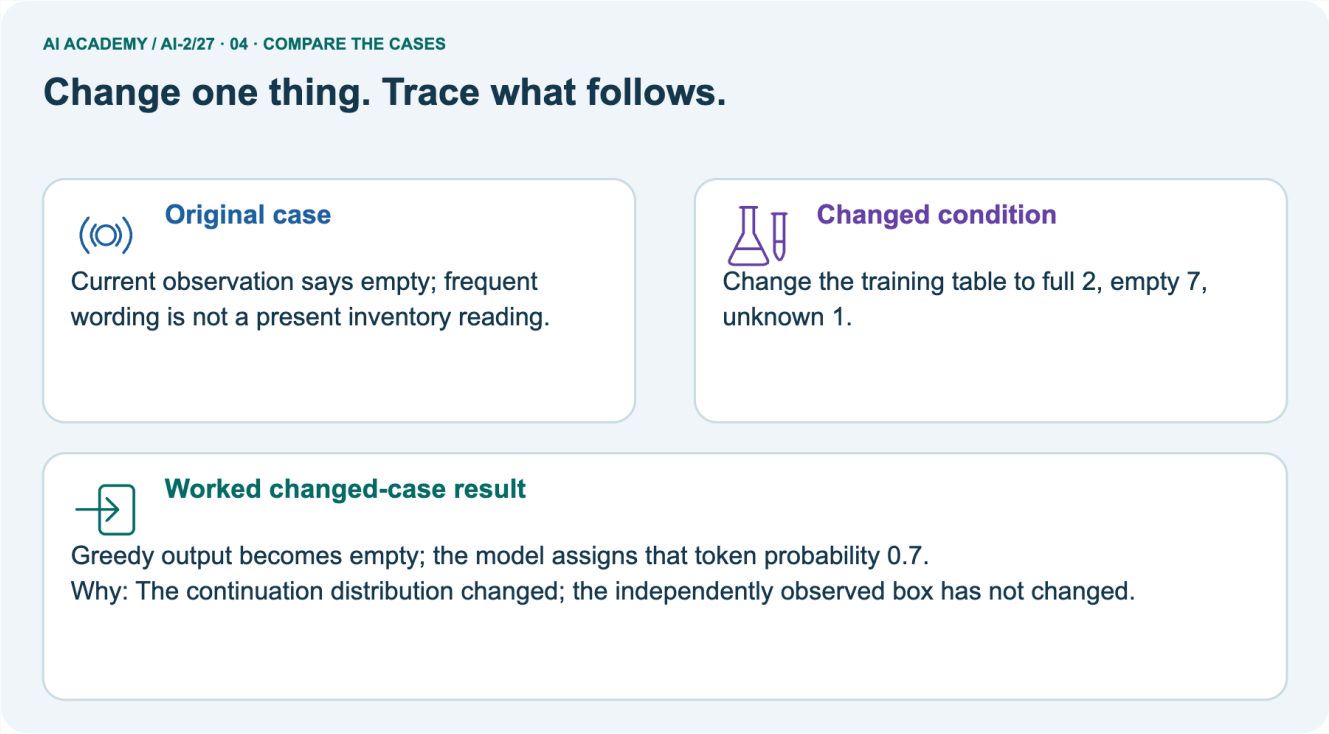
Toy next-word distribution. Units: probability. full: 0.6. empty: 0.3. unknown: 0.1.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

## 04. Change one thing. Trace what follows.




Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Current **observation** says empty; frequent wording is not a present inventory reading. Changed condition: Change the **training** table to full 2, empty 7, unknown 1.

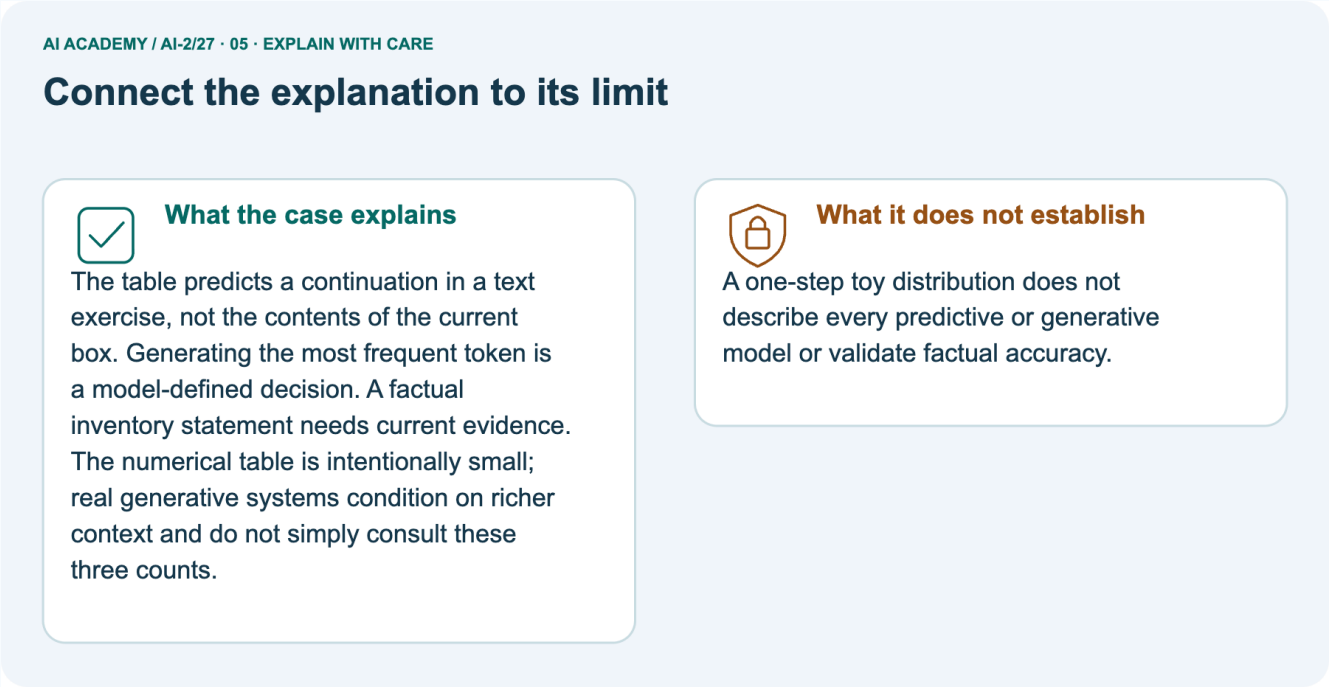
Expected result: Greedy **output** becomes empty; the model assigns that token probability 0.7. Reason: The continuation distribution changed; the independently observed box has not changed.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The continuation distribution changed; the independently observed box has not changed.

## 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

The table predicts a continuation in a text exercise, not the contents of the current box. Generating the most frequent token is a model-defined decision. A factual inventory statement needs current **evidence**.

The numerical table is intentionally small; real **generative** systems condition on richer context and do not simply consult these three counts.

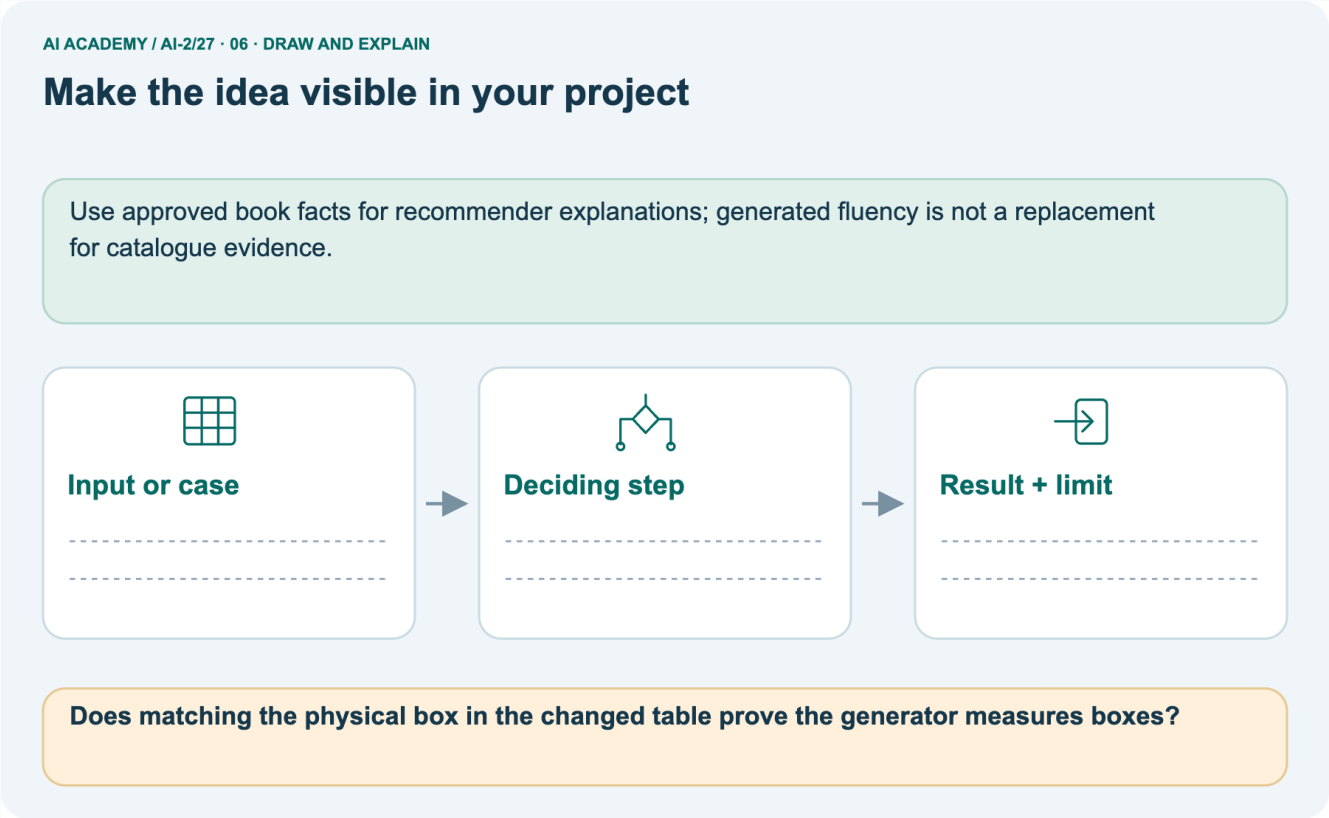
Limit: A one-step toy distribution does not describe every **predictive** or **generative** model or validate factual accuracy.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: A one-step toy distribution does not describe every **predictive** or **generative** model or validate factual accuracy.

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

Use approved book facts for recommender explanations; generated fluency is not a replacement for catalogue **evidence**. Draw your **input**, deciding step, result and limit.

Supported question: Does matching the physical box in the changed table prove the generator measures boxes?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: No. It happens to match here; its **mechanism** still selects a token from supplied counts rather than sensing contents.

## ▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

### 01. Predictive vs Generative: the case

### 02. Follow the deciding steps

### 03. Toy next-word distribution

### 04. Change one thing. Trace what follows.

### 05. Connect the explanation to its limit

### 06. Make the idea visible in your project

## ▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.